



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Fundamentals of Data Science

DS-503-MJ

Interview Preparation — Topic-wise Q&A

Semester I · Core

Curated interview questions and answers for Fundamentals of Data Science, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Core concepts

Q. What are the 3 V's of Big Data?

Volume (amount), Velocity (speed of generation) and Variety (structured, semi-structured, unstructured). Some add Veracity and Value.

Q. Describe the data science lifecycle.

Problem definition !' data collection !' cleaning/wrangling !' exploratory analysis !' modelling !' evaluation !' deployment !' communication.

Q. Structured vs unstructured data?

Structured data fits tables (rows/columns, e.g. SQL); unstructured data has no fixed schema (text, images, audio).

Preprocessing & EDA

Q. What is data wrangling?

Cleaning and transforming raw data into a usable form — handling missing values, outliers, inconsistent formats and encoding.

Q. How do you handle missing values?

Remove rows/columns, or impute with mean/median/mode, forward-fill, or model-based imputation, depending on the amount and pattern of missingness.

Q. Name distance measures for numeric data.

Euclidean, Manhattan and Minkowski distances; used in clustering and KNN.

Q. What is an outlier and how to detect it?

An observation far from others. Detect via boxplots/IQR, z-score, or model-based methods; handle by removal, capping or transformation.

